

Supplementary Material: *Shape calculus and automatic differentiation for multi-phase level-set topology optimisation with unfitted finite elements*

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1 Derivation of directional derivatives in Section 3.6

In the following, we derive the directional derivatives for the example in Section 3.6. As noted in Remark 3.22, the derivation is merely formal because it assumes shape differentiability of u_1 and u_2 . We define the Lagrangian $\mathcal{L}$ to be

$$\mathcal{L}(P, \Lambda; \phi_1, \phi_2) = F(P, P; \phi_1, \phi_2) - A(P, \Lambda; \phi_1, \phi_2) + l(\Lambda; \phi_1, \phi_2), \quad (\text{S.1})$$

where $P = [p_1, p_2]$ and $\Lambda = [\lambda_1, \lambda_2]$.

Since the Lagrangian is linear in Λ , differentiation of the Lagrangian with respect to Λ in the direction W generates the state equation residual, that is,

$$d_\Lambda \mathcal{L}(P, \Lambda; \phi_1, \phi_2)(W) \equiv \frac{d}{dt} \mathcal{L}(P, \Lambda + tW; \phi_1, \phi_2) \Big|_{t=0} = -A(P, W; \phi_1, \phi_2) + l(W; \phi_1, \phi_2), \quad (\text{S.2})$$

which vanishes for each $W \in \mathcal{V}_{1,h} \times \mathcal{V}_{2,h}$ when $P = U$ due to state equation (84). Next, differentiation of the Lagrangian with respect to P in the direction R generates a residual of the adjoint equation, that is,

$$\begin{aligned} d_P \mathcal{L}(P, \Lambda; \phi_1, \phi_2)(R) &\equiv \frac{d}{dt} \mathcal{L}(P + tR, \Lambda; \phi_1, \phi_2) \Big|_{t=0} \\ &= d_P F(P, P; \phi_1, \phi_2)(R) - d_P A(P, \Lambda; \phi_1, \phi_2)(R) = 2F(P, R; \phi_1, \phi_2) - A(R, \Lambda; \phi_1, \phi_2). \end{aligned} \quad (\text{S.3})$$

Finally, under the assumption that U is shape differentiable, the directional derivative of J in the direction w is obtained by differentiating the Lagrangian with respect to ϕ_i in the direction w and evaluating the resulting expression for $P = U$ and $\Lambda = \hat{\Lambda}$, that is,

$$\begin{aligned} d_{\phi_i} J(\phi_1, \phi_2)(w) &= d_{\phi_i} \mathcal{L}(P, \Lambda; \phi_1, \phi_2) \Big|_{P=U, \Lambda=\hat{\Lambda}}(w) \\ &= d_{\phi_i} F(U, U; \phi_1, \phi_2)(w) - d_{\phi_i} A(U, \hat{\Lambda}; \phi_1, \phi_2)(w) + d_{\phi_i} l(\hat{\Lambda}; \phi_1, \phi_2)(w). \end{aligned} \quad (\text{S.4})$$

where $\hat{\Lambda}$ satisfies the adjoint problem: find $\hat{\Lambda} \in \mathcal{V}_{1,h} \times \mathcal{V}_{2,h}$ such that

$$A(R, \hat{\Lambda}; \phi_1, \phi_2) = 2F(U, R; \phi_1, \phi_2) \quad \forall R \in \mathcal{V}_{1,h} \times \mathcal{V}_{2,h}. \quad (\text{S.5})$$

We now consider derivatives of the expressions $d_{\phi_i} F(U, U; \phi_1, \phi_2)(w)$, $-d_{\phi_i} A(U, \hat{\Lambda}; \phi_1, \phi_2)(w)$, and $d_{\phi_i} l(\hat{\Lambda}; \phi_1, \phi_2)(w)$ occurring in (S.4) term by term. (Note that these, by the scheme laid out in Section 3.6, are partial derivatives, where U and $\hat{\Lambda}$ are considered constant.) First, for the terms $-(\mathbf{A}_1 \nabla u_1, \nabla \lambda_1)_{D_{\phi_1} \cap \overline{D}_{\phi_2}^c} - (\mathbf{A}_2 \nabla u_2, \nabla \lambda_2)_{D_{\phi_1} \cap D_{\phi_2}}$ that appear in $A(U, \hat{\Lambda}; \phi_1, \phi_2)$, we obtain, using Theorem 3.10, that

$$\begin{aligned} d_{\phi_1} \left[-(\mathbf{A}_1 \nabla u_1, \nabla \hat{\lambda}_1)_{D_{\phi_1} \cap \overline{D}_{\phi_2}^c} - (\mathbf{A}_2 \nabla u_2, \nabla \hat{\lambda}_2)_{D_{\phi_1} \cap D_{\phi_2}} \right] (w) \\ = \int_{\partial D_{\phi_1} \cap \overline{D}_{\phi_2}^c} (\mathbf{A}_1 \nabla u_1) \cdot \nabla \hat{\lambda}_1 \frac{w}{|\partial_{\mathbf{n}_{\phi_1}} \phi_1|} \, dS + \int_{\partial D_{\phi_1} \cap D_{\phi_2}} (\mathbf{A}_2 \nabla u_2) \cdot \nabla \hat{\lambda}_2 \frac{w}{|\partial_{\mathbf{n}_{\phi_1}} \phi_1|} \, dS, \end{aligned} \quad (\text{S.6})$$

$$\begin{aligned}
d_{\phi_2} \left[-(\mathbf{A}_1 \nabla u_1, \nabla \hat{\lambda}_1)_{D_{\phi_1} \cap \bar{D}_{\phi_2}^c} - (\mathbf{A}_2 \nabla u_2, \nabla \hat{\lambda}_2)_{D_{\phi_1} \cap D_{\phi_2}} \right] (w) \\
= - \int_{D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{A}_1 \nabla u_1) \cdot \nabla \hat{\lambda}_1 \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS + \int_{D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{A}_2 \nabla u_2) \cdot \nabla \hat{\lambda}_2 \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS,
\end{aligned} \tag{S.7}$$

and, similarly,

$$\begin{aligned}
d_{\phi_1} F(U, U; \phi_1, \phi_2)(w) \\
= d_{\phi_1} \left[(\mathbf{A}_1 \nabla u_1, \nabla u_1)_{D_{\phi_1} \cap \bar{D}_{\phi_2}^c} + (\mathbf{A}_2 \nabla u_2, \nabla u_2)_{D_{\phi_1} \cap D_{\phi_2}} \right] (w) \\
= - \int_{\partial D_{\phi_1} \cap \bar{D}_{\phi_2}^c} (\mathbf{A}_1 \nabla u_1) \cdot \nabla u_1 \frac{w}{|\partial \mathbf{n}_{\phi_1} \phi_1|} dS - \int_{\partial D_{\phi_1} \cap D_{\phi_2}} (\mathbf{A}_2 \nabla u_2) \cdot \nabla u_2 \frac{w}{|\partial \mathbf{n}_{\phi_1} \phi_1|} dS,
\end{aligned} \tag{S.8}$$

$$\begin{aligned}
d_{\phi_2} F(U, U; \phi_1, \phi_2)(w) \\
= d_{\phi_2} \left[(\mathbf{A}_1 \nabla u_1, \nabla u_1)_{D_{\phi_1} \cap \bar{D}_{\phi_2}^c} + (\mathbf{A}_2 \nabla u_2, \nabla u_2)_{D_{\phi_1} \cap D_{\phi_2}} \right] (w) \\
= \int_{D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{A}_1 \nabla u_1) \cdot \nabla u_1 \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS - \int_{D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{A}_2 \nabla u_2) \cdot \nabla u_2 \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS,
\end{aligned} \tag{S.9}$$

In this example, we will consider the heat source boundary Γ_N fixed, independent of the design. Therefore, trivially,

$$d_{\phi_i} l(\hat{\Lambda}; \phi_1, \phi_2)(w) = 0. \tag{S.10}$$

Also, since perturbations of ϕ_1 and ϕ_2 satisfy Assumption 3.5,

$$d_{\phi_i} j(V, U) = 0. \tag{S.11}$$

Next, we consider the directional derivatives of the interface terms in $-A(U, \hat{\Lambda}; \phi_1, \phi_2)$, that is,

$$(\{\mathbf{A} \nabla u\}, [\hat{\lambda}])_{D_{\phi_1} \cap \partial D_{\phi_2}} + (\{\mathbf{A} \nabla \hat{\lambda}\}, [u])_{D_{\phi_1} \cap \partial D_{\phi_2}} - (\mu [\hat{\lambda}], [u])_{D_{\phi_1} \cap \partial D_{\phi_2}}. \tag{S.12}$$

Recall that there are two different results for differentiation of boundary integrals, Theorems 3.15 and 3.18. The former concerns integrals that are truncated by a domain that is perturbed by the level-set function. (For an illustration, consider differentiating integrals over Γ_{12} under a perturbation of ϕ_1 in Figure 1.) This is precisely the case when differentiating the interface terms (S.12) under a perturbation of ϕ_1 . Using Theorem 3.15, we thus have

$$\begin{aligned}
d_{\phi_1} \left[(\{\mathbf{A} \nabla u\}, [\hat{\lambda}])_{D_{\phi_1} \cap \partial D_{\phi_2}} + (\{\mathbf{A} \nabla \hat{\lambda}\}, [u])_{D_{\phi_1} \cap \partial D_{\phi_2}} - (\mu [\hat{\lambda}], [u])_{D_{\phi_1} \cap \partial D_{\phi_2}} \right] (w) \\
= \int_{\partial D_{\phi_1} \cap \partial D_{\phi_2}} \left(\{\mathbf{A} \nabla u\} \cdot [\hat{\lambda}] + \{\mathbf{A} \nabla \hat{\lambda}\} \cdot [u] - \mu [u] [\hat{\lambda}] \right) \frac{w}{|\partial \mathbf{N}^2 \phi_1|} d\gamma,
\end{aligned} \tag{S.13}$$

where $\mathbf{N}^2$ is the unit normal to ∂D_{ϕ_1} inside ∂D_{ϕ_2} , outward with respect to D_{ϕ_2} .

In contrast, the differentiation of interface terms (S.12) with respect to ϕ_2 concerns a boundary given by a level-set function that itself is perturbed, a case that is covered by Theorem 3.18. (For an illustration, consider differentiating integrals over Γ_{12} under a perturbation of ϕ_2 in Figure 1.) Moreover, both the average and jump operators in expression (S.12) involve the normal field, which in turn depends on ϕ_2 . We will thus require the directional derivative of the outward-directed normal field $\mathbf{n}_{\phi_2}$ on ∂D_{ϕ_2} . As shown by Berggren et al. (2026, expression (B18)), the derivative of $\mathbf{n}_{\phi_2}$ with respect to ϕ_2 in the direction w is given by

$$d_{\phi_2} \mathbf{n}_{\phi_2}(w) = \frac{P_T \nabla w}{|\partial \mathbf{n}_{\phi_2} \phi_2|}, \tag{S.14}$$

where $P_T = \mathbf{I} - \mathbf{n}_{\phi_2} \otimes \mathbf{n}_{\phi_2}$ is the projector on to the tangent plane of ∂D_{ϕ_2} . Using this result, Theorem 3.18, and

the product rule as discussed in Remark 3.21, we have, using notation further elaborated on below,

$$\begin{aligned}
& d_{\phi_2} \left[(\{\mathbf{A} \nabla u\}, [\hat{\lambda}])_{D_{\phi_1} \cap \partial D_{\phi_2}} + (\{\mathbf{A} \nabla \hat{\lambda}\}, [u])_{D_{\phi_1} \cap \partial D_{\phi_2}} - (\mu[\hat{\lambda}], [u])_{D_{\phi_1} \cap \partial D_{\phi_2}} \right] (w) \\
&= \int_{D_{\phi_1} \cap \partial D_{\phi_2}} d_{\phi_2} \{\mathbf{A} \nabla u\}(w) \cdot [\hat{\lambda}] + d_{\phi_2} \{\mathbf{A} \nabla \hat{\lambda}\}(w) \cdot [u] \, dS \\
&+ \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \{\mathbf{A} \nabla u\} \cdot d_{\phi_2} [\hat{\lambda}](w) + \{\mathbf{A} \nabla \hat{\lambda}\} \cdot d_{\phi_2} [u](w) \, dS \\
&- \mu \int_{D_{\phi_1} \cap \partial D_{\phi_2}} d_{\phi_2} [\hat{\lambda}](w) \cdot [u] + [\hat{\lambda}] \cdot d_{\phi_2} [u](w) \, dS \\
&- \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \left(\{\mathbf{A} \partial_{\mathbf{n}_{\phi_2}} \nabla u\} [\hat{\lambda}] + \{\mathbf{A} \nabla u\} [\partial_{\mathbf{n}_{\phi_2}} \hat{\lambda}] \right) \frac{w}{|\partial_{\mathbf{n}_{\phi_2}} \phi_2|} \, dS \\
&- \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \left(\{\mathbf{A} \partial_{\mathbf{n}_{\phi_2}} \nabla \hat{\lambda}\} [u] + \{\mathbf{A} \nabla \hat{\lambda}\} [\partial_{\mathbf{n}_{\phi_2}} u] \right) \frac{w}{|\partial_{\mathbf{n}_{\phi_2}} \phi_2|} \, dS \\
&+ \mu \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \left([\partial_{\mathbf{n}_{\phi_2}} \hat{\lambda}] [u] + [\hat{\lambda}] [\partial_{\mathbf{n}_{\phi_2}} u] \right) \frac{w}{|\partial_{\mathbf{n}_{\phi_2}} \phi_2|} \, dS \\
&+ \sum_{S \in \mathcal{F}_h} \int_{S \cap D_{\phi_1} \cap \partial D_{\phi_2}} \left[\mathbf{n}_{\phi_2} \left(\{\mathbf{A} \nabla \hat{\lambda}\} \cdot [u] + \{\mathbf{A} \nabla u\} \cdot [\hat{\lambda}] - \mu[u] \cdot [\hat{\lambda}] \right) \right]_S \frac{w}{|\partial_{\mathbf{N}^S} \phi_2|} \, d\gamma \\
&+ \int_{\partial D \cap D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{n}_{\partial D} \cdot \mathbf{n}_{\phi_2}) \left(\{\mathbf{A} \nabla \hat{\lambda}\} \cdot [u] + \{\mathbf{A} \nabla u\} \cdot [\hat{\lambda}] - \mu[u] \cdot [\hat{\lambda}] \right) \frac{w}{|\partial_{\mathbf{N}^D} \phi_2|} \, d\gamma \\
&+ \int_{\partial D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{n}_{\phi_1} \cdot \mathbf{n}_{\phi_2}) \left(\{\mathbf{A} \nabla \hat{\lambda}\} \cdot [u] + \{\mathbf{A} \nabla u\} \cdot [\hat{\lambda}] - \mu[u] \cdot [\hat{\lambda}] \right) \frac{w}{|\partial_{\mathbf{N}^1} \phi_1|} \, d\gamma,
\end{aligned} \tag{S.15}$$

where the three first integrals on the right side correspond to the the g' term in expression (81), whereas the rest of the terms correspond to the $dJ(\phi_k)(w)$ term. Regarding the first integral on the right, for f either u or $\hat{\lambda}$, it holds that

$$d_{\phi_2} \{\mathbf{A} \nabla f\}(w) = d_{\phi_2} (\kappa_1 \mathbf{A}_1 \nabla f_1 + \kappa_2 \mathbf{A}_2 \nabla f_2)(w) = (d_{\phi_2} \kappa_1(w)) \mathbf{A}_1 \nabla f_1 + (d_{\phi_2} \kappa_2(w)) \mathbf{A}_2 \nabla f_1, \tag{S.16}$$

where, by differentiation, utilising definition (86) and formula (S.14) ,

$$\begin{aligned}
d_{\phi_2} \kappa_1(w) &= 2\mathbf{n}_{\phi_2} \cdot \frac{c_1 \mathbf{A}_2 - c_2 \mathbf{A}_1}{(c_1 + c_2)^2} d_{\phi_2} \mathbf{n}_{\phi_2}(w) = 2\mathbf{n}_{\phi_2} \cdot \frac{(c_1 \mathbf{A}_2 - c_2 \mathbf{A}_1)}{(c_1 + c_2)^2} \frac{P_T \nabla w}{|\partial_{\mathbf{n}_{\phi_2}} \phi_2|}, \\
d_{\phi_2} \kappa_2(w) &= 2\mathbf{n}_{\phi_2} \cdot \frac{c_2 \mathbf{A}_1 - c_1 \mathbf{A}_2}{(c_1 + c_2)^2} d_{\phi_2} \mathbf{n}_{\phi_2}(w) = 2\mathbf{n}_{\phi_2} \cdot \frac{(c_2 \mathbf{A}_1 - c_1 \mathbf{A}_2)}{(c_1 + c_2)^2} \frac{P_T \nabla w}{|\partial_{\mathbf{n}_{\phi_2}} \phi_2|}
\end{aligned} \tag{S.17}$$

Moreover, the derivatives of the jumps in the the second and third integrals can be written, where, again, f is either u or $\hat{\lambda}$,

$$\begin{aligned}
d_{\phi_2} [f](w) &= d_{\phi_2} (f_1 \mathbf{n}_1 + f_2 \mathbf{n}_2)(w) = f_1 d_{\phi_2} \mathbf{n}_1(w) + f_2 d_{\phi_2} \mathbf{n}_2(w) \\
&= f_1 \mathbf{n}_1 \cdot \mathbf{n}_1 d_{\phi_2} \mathbf{n}_1(w) + f_2 \mathbf{n}_2 \cdot \mathbf{n}_2 d_{\phi_2} \mathbf{n}_2(w) \\
&= f_1 \mathbf{n}_1 \cdot \mathbf{n}_{\phi_2} d_{\phi_2} \mathbf{n}_{\phi_2}(w) + f_2 \mathbf{n}_2 \cdot \mathbf{n}_{\phi_2} d_{\phi_2} \mathbf{n}_{\phi_2}(w) \\
&= [f \mathbf{n}_{\phi_2}] d_{\phi_2} \mathbf{n}_{\phi_2}(w) = [f \mathbf{n}_{\phi_2}] \frac{P_T \nabla w}{|\partial_{\mathbf{n}_{\phi_2}} \phi_2|}
\end{aligned} \tag{S.18}$$

Finally, since we employ piecewise-linear approximations, the first term in the fourth and fifth integrals vanish here, since they involve second derivatives inside the elements.

Bringing everything together, we find, by substituting derivatives (S.8), (S.6), (S.13), (S.10), and (S.11) into expression (S.4), that

$$\begin{aligned}
d_{\phi_1} J(\phi_1, \phi_2)(w) &= \int_{\partial D_{\phi_1} \cap \overline{D}_{\phi_2}^c} (\mathbf{A}_1 \nabla u_1) \cdot (\nabla \hat{\lambda}_1 - \nabla u_1) \frac{w}{|\partial_{\mathbf{n}_{\phi_1}} \phi_1|} \, dS \\
&+ \int_{\partial D_{\phi_1} \cap D_{\phi_2}} (\mathbf{A}_2 \nabla u_2) \cdot (\nabla \hat{\lambda}_2 - \nabla u_2) \frac{w}{|\partial_{\mathbf{n}_{\phi_1}} \phi_1|} \, dS \\
&+ \int_{\partial D_{\phi_1} \cap \partial D_{\phi_2}} \left(\{\mathbf{A} \nabla u\} \cdot [\hat{\lambda}] + \{\mathbf{A} \nabla \hat{\lambda}\} \cdot [u] - \mu[u] \cdot [\hat{\lambda}] \right) \frac{w}{|\partial_{\mathbf{N}^2} \phi_1|} \, d\gamma,
\end{aligned} \tag{S.19}$$

and, by combining derivatives (S.9), (S.7), (S.15), (S.10), and (S.11) into expression (S.4), that

$$\begin{aligned}
d_{\phi_2} J(\phi_1, \phi_2)(w) = & - \int_{D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{A}_1 \nabla u_1) \cdot (\nabla \hat{\lambda}_1 - \nabla u_1) \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS \\
& + \int_{D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{A}_2 \nabla u_2) \cdot (\nabla \hat{\lambda}_2 - \nabla u_2) \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS \\
& + \int_{D_{\phi_1} \cap \partial D_{\phi_2}} d_{\phi_2} \{ \mathbf{A} \nabla u \}(w) \cdot \llbracket \hat{\lambda} \rrbracket + d_{\phi_2} \{ \mathbf{A} \nabla \hat{\lambda} \}(w) \cdot \llbracket u \rrbracket dS \\
& + \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \{ \mathbf{A} \nabla u \} \cdot d_{\phi_2} \llbracket \hat{\lambda} \rrbracket(w) + \{ \mathbf{A} \nabla \hat{\lambda} \} \cdot d_{\phi_2} \llbracket u \rrbracket(w) dS \\
& - \mu \int_{D_{\phi_1} \cap \partial D_{\phi_2}} d_{\phi_2} \llbracket \hat{\lambda} \rrbracket(w) \cdot \llbracket u \rrbracket + \llbracket \hat{\lambda} \rrbracket \cdot d_{\phi_2} \llbracket u \rrbracket(w) dS \\
& - \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \left(\{ \mathbf{A} \partial_{\mathbf{n}_{\phi_2}} \nabla u \} \llbracket \hat{\lambda} \rrbracket + \{ \mathbf{A} \nabla u \} \llbracket \partial_{\mathbf{n}_{\phi_2}} \hat{\lambda} \rrbracket \right) \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS \\
& - \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \left(\{ \mathbf{A} \partial_{\mathbf{n}_{\phi_2}} \nabla \hat{\lambda} \} \llbracket u \rrbracket + \{ \mathbf{A} \nabla \hat{\lambda} \} \llbracket \partial_{\mathbf{n}_{\phi_2}} u \rrbracket \right) \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS \\
& + \mu \int_{D_{\phi_1} \cap \partial D_{\phi_2}} \left(\llbracket \partial_{\mathbf{n}_{\phi_2}} \hat{\lambda} \rrbracket \llbracket u \rrbracket + \llbracket \hat{\lambda} \rrbracket \llbracket \partial_{\mathbf{n}_{\phi_2}} u \rrbracket \right) \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} dS \\
& + \sum_{S \in \mathcal{F}_h \setminus \partial D} \int_{S \cap D_{\phi_1} \cap \partial D_{\phi_2}} \left[\mathbf{n}_{\phi_2} \left(\{ \mathbf{A} \nabla \hat{\lambda} \} \cdot \llbracket u \rrbracket + \{ \mathbf{A} \nabla u \} \cdot \llbracket \hat{\lambda} \rrbracket - \mu \llbracket u \rrbracket \cdot \llbracket \hat{\lambda} \rrbracket \right) \right] \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} d\gamma \\
& + \int_{\partial D \cap D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{n}_{\partial D} \cdot \mathbf{n}_{\phi_2}) \left(\{ \mathbf{A} \nabla \hat{\lambda} \} \cdot \llbracket u \rrbracket + \{ \mathbf{A} \nabla u \} \cdot \llbracket \hat{\lambda} \rrbracket - \mu \llbracket u \rrbracket \cdot \llbracket \hat{\lambda} \rrbracket \right) \frac{w}{|\partial \mathbf{n}_{\phi_2} \phi_2|} d\gamma \\
& + \int_{\partial D_{\phi_1} \cap \partial D_{\phi_2}} (\mathbf{n}_{\phi_1} \cdot \mathbf{n}_{\phi_2}) \left(\{ \mathbf{A} \nabla \hat{\lambda} \} \cdot \llbracket u \rrbracket + \{ \mathbf{A} \nabla u \} \cdot \llbracket \hat{\lambda} \rrbracket - \mu \llbracket u \rrbracket \cdot \llbracket \hat{\lambda} \rrbracket \right) \frac{w}{|\partial \mathbf{n}_{\phi_1} \phi_1|} d\gamma,
\end{aligned} \tag{S.20}$$

where $\hat{\Lambda} = [\hat{\lambda}_1, \hat{\lambda}_2]$ satisfies the adjoint problem: Find $\hat{\Lambda} \in \mathcal{V}_{1,h} \times \mathcal{V}_{2,h}$ such that

$$\begin{aligned}
& (\mathbf{A}_1 \nabla w_1, \nabla \hat{\lambda}_1)_{D_{\phi_1} \cap \overline{D}_{\phi_2}^c} + (\mathbf{A}_2 \nabla w_2, \nabla \hat{\lambda}_2)_{D_{\phi_1} \cap D_{\phi_2}} + j(W, \hat{\Lambda}) + (\mu \llbracket \hat{\lambda} \rrbracket, \llbracket W \rrbracket)_{D_{\phi_1} \cap \partial D_{\phi_2}} \\
& - (\{ \mathbf{A} \nabla \hat{\lambda} \}, \llbracket W \rrbracket)_{D_{\phi_1} \cap \partial D_{\phi_2}} - (\{ \mathbf{A} \nabla W \}, \llbracket \hat{\lambda} \rrbracket)_{D_{\phi_1} \cap \partial D_{\phi_2}} = 2F(W, \phi_1, \phi_2),
\end{aligned} \tag{S.21}$$

for all $W \in \mathcal{V}_{1,h} \times \mathcal{V}_{2,h}$.

Details on the evaluations of the various directional derivatives inside reduced gradient (S.20) are given in expressions (S.16), (S.17), and (S.18). Note also that the first term in integrals six and seven in expression (S.20) vanishes in this case, since we use piecewise-linear approximation spaces.

References

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